

Statistical Process Control in Semiconductor Manufacturing

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The fabrication of integrated circuits (IC's) has always promoted technical innovation. IC research and development, however, has traditionally been focused on improving the performance of the product, while paying relatively little attention to the efficiency of its production. Today, standard scientific manufacturing practices are finally finding their way into IC production. In most cases, the development of an industrial mentality coincides with the introduction of statistical process control (SPC). This paper undertakes a brief survey of standard SPC schemes, and illustrates them through examples taken from the semiconductor industry. These methods range from contamination control to the monitoring of continuous process parameters. Even as SPC is transforming IC production, the peculiarities of semiconductor manufacturing technology are transforming SPC. Therefore, the second part of this paper describes novel SPC applications which are now emerging in semiconductor production. These methods are being developed to monitor the short production runs that are characteristic of flexible manufacturing. Additional SPC techniques suitable for in situ multivariate sensor readings are also discussed.

I. INTRODUCTION

Historians divide the various evolutionary stages of modern manufacturing practice into six main periods [1], [2]. Throughout each of these periods, the objectives of efficiency and profitability have been pursued by focusing on different aspects of production. During the early 1800's, manufacturing practice was revolutionized by the introduction of machine tools that could achieve unprecedented mechanical accuracy. In the 1850's, the era of efficient mass production opened with the introduction of interchangeable parts. In the 1900's, Taylor introduced the concept of the scientific management of labor [3]. Around 1930, Walter Shewhart opened a new era by introducing statistical process control (SPC). The availability of computers made possible the introduction of the numerical control technologies that ushered us into the age of automation

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in the 1970's. All these advances finally culminated in the computer integrated manufacturing (CIM) systems of the 1980's.

While other industrial sectors had over two hundred years to absorb these changes, semiconductor manufacturing was thrust through them in less than two short decades. This transition has not been smooth and it is not yet complete. Nevertheless, powerful economic forces are finally transforming the laboratory art of IC fabrication into the modern science of manufacturing. Even though the cleanroom has traditionally been the domain of technology researchers and experimenters, the teachings of scientific manufacturing and the science of high volume production are finally being applied there. The subject of this paper is the application of SPC in modern semiconductor production.

SPC has been widely recognized as a tool that had technical as well as cultural impact on production [4]. Since its introduction in the 1930's by Walter Shewhart, SPC transformed the principles of production by transferring responsibility directly to the factory floor operator. This was accomplished by introducing a simple, yet powerful tool, the control chart. The control chart can be used as a gauge to detect and thus help eliminate unnecessary sources of variability. The inherent simplicity of these early control procedures empowered the operators with important decisions right at the point of production.

Since the 1930's the technical contributions to SPC have grown tremendously, in order to keep in pace with the advancing technology. Soon after its introduction, Shewhart's simple chart had to be augmented to accommodate various distributions, errors in measurements, small drifts as well as abrupt shifts, cyclic maintenance patterns, etc. These requirements led to the introduction of a large number of control procedures. Although some of the modern control techniques require substantial computational effort, relatively inexpensive computing hardware nonetheless enables the operator to use them at the point of production.

Today, the technology of production has evolved to the point, that new and complex SPC methods are necessary; in response, methods like multivariate statistics, time series modeling, intelligent control charts, etc., are gaining wider acceptance. These advantages are coming after a

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host of statistical production techniques became popular in the semiconductor manufacturing industry during the late eighties [6]–[9].

The objective of this paper is to give a technical overview of the application and impact of SPC in semiconductor manufacturing. This paper consists of two major parts: the first is a short overview of the application of traditional SPC concepts, and the second is a summary of some of the modern SPC techniques that are finding use in semiconductor production.

Because of SPC's wide impact in the culture of production, it has been observed, and rightly so, that the introduction of SPC is as much a managerial challenge as it is a technical one. This paper will address the technical side of this issue.

II. SPC—BASIC CONCEPTS

SPC was introduced in the early 1930's by Walter Shewhart of Bell Telephone and Telegraph [9]. Shewhart's original objective was to provide a simple, intuitive way to summarize the history of the process. This summary was to serve as a reference to gauge present production performance. If future performance was found to be significantly different from its historical norm, then an alarm would be issued. By flagging significant process deviations and by finding and correcting their causes, the quality of a manufacturing process was improved.

More specifically, a process is said to be in *statistical control* when it displays nothing but the routine run by run variation. Unusual patterns and departures from that variation are indications that the process is *out of statistical control*. This implies that the process is experiencing a change that cannot be dismissed as routine variation. A central idea in SPC is that of the existence of an *assignable* (or "*special*") *cause* behind any significant deviation from the historical norm. In other words, an assignable cause is the reason behind every true alarm. The term "assignable" implies something that can be discovered and corrected, such as a chamber door that does not seal properly, a contaminated gas line, a miscalibrated film thickness meter, incoming wafers that are out of specifications, operator errors, etc. An assignable cause is to be contrasted to *chronic* or *routine* sources of variation, such as measurement errors caused by the inherent lack of precision in a photospectrometer, or the variation in the resistivity of a deposited layer, caused by the limited precision of the mass flow controllers in a furnace.

In general, *common causes* are the reason behind the imperfect run to run repeatability of a process. This repeatability is limited by the precision of the manufacturing equipment, the routine variation of the incoming material, the environmental cleanroom controls, etc. By definition, the operator cannot remove common causes from the process.¹

The role of an SPC procedure is then the formalization of the decision as to whether the process is operating under

¹Common causes can, however, be controlled by management actions, such as instituting tighter specifications on incoming material, retraining the operators, and upgrading critical pieces of equipment.

statistical control. From a statistical point of view, SPC is a formal *hypothesis test*. This test will objectively choose between two hypotheses. The first hypothesis, known as the *null hypothesis* (H_0) is that the process is under statistical control. This assertion implies that there are no assignable causes of variation and that therefore the process is operating as consistently as possible. The second hypothesis, known as the *alternate hypothesis* (H_a) is that the process is out of statistical control. This implies that an assignable cause is present, and that this assignable cause should be discovered and removed in order to regain control of the process.

There are many ways that this hypothesis test is actually implemented. In its simplest form, the test consists of plotting one performance parameter against an upper and a lower control limit. These limits have been set in order to reflect the past behavior of the process. If these limits are met, then we accept H_0 and we declare that the process is under statistical control. If H_0 is rejected, then we adopt the alternate hypothesis H_a that stipulates the existence of an assignable cause. In this case, a misprocessing alarm is issued to the operator.

Naturally, it is essential that the alarms are reliably generated in the presence of the day to day variation which is characteristic of high volume production. Unfortunately, any statistical test which operates on a limited set of data is subject to errors. There are two types of errors that can be made while choosing among the two hypotheses. The first, known as a *type I error*, is to mistakenly reject H_0 , or, equivalently, to issue a false alarm. The second error, known as a *type II error*, to mistakenly accept H_a , or, equivalently, to miss issuing an alarm. The probability of committing a type I error is also known as the *manufacturer's risk*, since it leads to unnecessary disruptions during production. The probability of a type II error is the *consumer's risk*, since it leads to producing defective products.

Once the hypothesis test has been defined, it is possible to estimate the probabilities of committing either one of these errors. This will permit the fine tuning of the statistical test so that production costs incurred by faulty decisions will be minimized. The following discussions address the calculation of the types I and II error rates in the context of the basic control chart.

A. The Control Chart and Examples of Its Application

In semiconductor manufacturing, more so than in other production areas, we often monitor a crucial process by means of measuring and recording one or more critical parameters. For example, the lithographic sequence can be monitored by, among other things, recording the thickness of the photoresist layer before the exposure. This layer is deposited by spinning a silicon wafer while the viscous photoresist solidifies. The target thickness of the photoresist is typically about 1 μm . Because of the mechanical nature of this operation, even if our equipment is functioning properly, the thickness of the layer on each wafer will vary. In addition, the measurement of the deposited layer,

usually done with a photospectrometer, will be subject to calibration errors. The combination of these sources of variability give us a photoresist thickness that, as recorded, appears to be statistically distributed with a routine run to run variation. The goal of SPC is to implement a simple procedure that will reliably flag any significant deviation beyond the routine run to run variation.

The control chart is a simple and effective graphical representation of the process status. It can also serve as a very basic implementation of the hypothesis test discussed in the previous section. There are many types of control charts, each suitable to a different application. We will first describe the simple $\bar{X}$ chart², as a vehicle for illustrating some fundamental SPC concepts, such as the *types I and II errors*, the *average run length*, and the *operating characteristic function*.

The $\bar{X}$ chart is based on the assumption that, when the process is in control, the monitored variable is distributed according to a normal distribution with a known mean μ and a known sigma σ , symbolically:

$$x_i \sim N(\mu, \sigma^2). \quad (1)$$

Using the $\bar{X}$ chart consists of grouping and averaging n readings of x , defined as

$$\bar{x} \equiv \frac{1}{n} \sum_{i=1}^n x_i. \quad (2)$$

Under the assumption that each reading is *Independently and Identically Normally Distributed* (thereafter to be known as the *IIND assumption*), the arithmetic average can be shown to be distributed according to another known distribution given as

$$\bar{x} \sim N\left(\mu, \frac{\sigma^2}{n}\right). \quad (3)$$

Pictorially, this control scheme is implemented by plotting the value of the group average versus time. If this value falls within a zone of high likelihood (determined by the known distribution of the plotted statistic), then we conclude that the process is under control. If the value falls outside this high likelihood region, the process is considered to be out of control. Traditionally, the high likelihood region is chosen to be within $\pm 3\sigma_{\bar{x}}$, where $\sigma_{\bar{x}}$ is the standard deviation of the arithmetic average. It can be shown that the *three-sigma* control limits yield a probability of type I error equal to 0.0027. These limits are defined as follows:

$$CL = \bar{x} \pm 3 \frac{\sigma}{\sqrt{n}}. \quad (4)$$

If a point falls outside this zone, then we can conclude at an *0.0027 level of significance*, that this point is now generated by a different distribution, one that has shifted its mean, its variance or both.

²In this document, the symbol $\bar{x}$ indicates the arithmetic average of the random variable x . Most statistical symbols are not consistent across the literature. Whenever possible, I have adopted the symbols used in [9].

As an example, consider the $\bar{X}$ chart that was implemented to control the thickness of photoresist in the Berkeley microfabrication laboratory. Assuming that we know the mean and variance of this chart, we plot the average thickness versus time in Fig. 1.

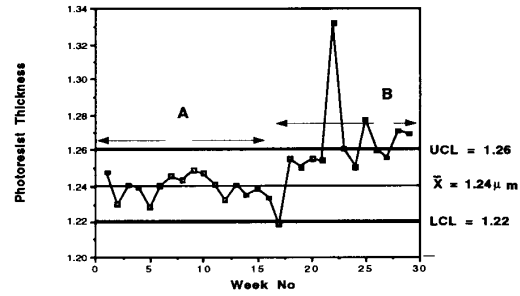


Fig. 1. $\bar{X}$ chart of photoresist thickness in the Berkeley Micro-fabrication Laboratory.

In addition to the types I and type II risks, the *average run length* (ARL) is also an important characteristic of the control chart. The ARL is defined as the average number of points plotted between alarms, false or otherwise. The ARL is a function of the process status and the type I and II risks. In Fig. 1, the process appears to be under control in the region designated as A. Here, we want the ARL to be as long as possible (because any generated alarms will be false), and in fact $ARL = 1/\alpha$. If $\alpha = 0.0027$ (for three sigma control limits), then the average number of samples between alarms is about 370.

Again in Fig. 1 the process appears to be out of control in the region designated as B. Here, we want the ARL to be as short as possible, and in fact the *out of control* ARL is equivalent to $1/(1 - \beta)$. In this region β , the type II risk, is related to the size and the type of the process shift, as illustrated in Fig. 2.

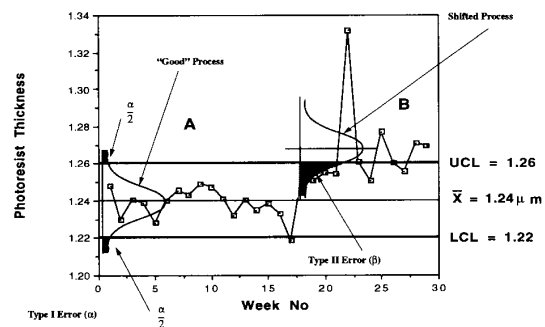


Fig. 2. Relations between type I and type II risks for an $\bar{X}$ chart.

Assuming that the process goes out of control because its mean has shifted while its variance stayed the same, it is possible to plot β versus k , the amount of the shift

(expressed in units of standard deviation) and the sample size n , which we employ in the calculation of the average. In Fig. 3, which has been adapted from [9], we show such a plot, known as the *operating characteristic function* of the chart. Next we discuss the application of some of the more sophisticated traditional SPC tools.

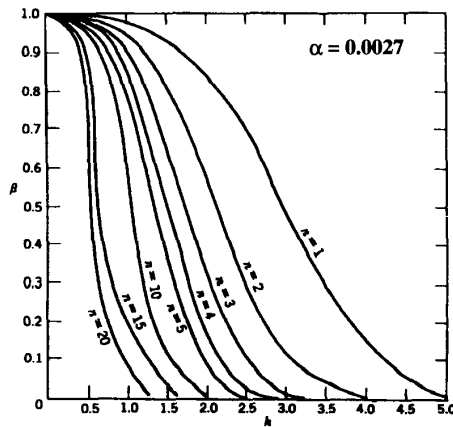


Fig. 3. The operating characteristic function of the $\bar{X}$ chart.

B. Controlling Location and Spread

In semiconductor manufacturing as well as in most manufacturing disciplines, production quality can be degraded by a shift of the mean or by an increase of the variance. For this reason, additional charts were created as companions to the $\bar{X}$ chart, in order to guard against an increase in the process spread. Two types of such charts are described below.

1) *The $\bar{X}$ -R Charts:* The simplest example of such a chart is the range (R) chart. The range of a group of measurements, defined as $R = x_{\max} - x_{\min}$, is a statistic with a known distribution [9]. This statistic is related to the standard deviation of the normal distribution that generated the measurements. More specifically, the average range $\bar{R}$ of m of groups can be shown to estimate the sigma of the distribution that generated the group of measurements:

$$\hat{\sigma} = \frac{\bar{R}}{d_2}. \quad (5)$$

Here d_2 is a proportionality factor which depends upon the size n of the group, and whose values have been tabulated in many standard textbooks of applied statistics. Furthermore, the standard deviation of the estimated range is also related to the standard deviation of the group:

$$\sigma_R = d_3 \sigma \quad (6)$$

where d_3 is a factor which depends on the size of the group and whose values have also been tabulated in standard statistical textbooks. Since σ is unknown, σ_R is estimated

by the following relationship:

$$\hat{\sigma}_R = d_3 \frac{\bar{R}}{d_2}. \quad (7)$$

Equations (5)–(7) form the basis for defining the three-sigma ($\alpha = 0.0027$) control limits for the $\bar{X}$ and the R charts:

$$CL_R = \bar{R} \pm 3d_3 \frac{\bar{R}}{d_2} \quad (8)$$

$$CL_{\bar{X}} = \bar{\bar{x}} \pm \frac{3}{d_2 \sqrt{n}} \bar{R} \quad (9)$$

where $\bar{R}$ is the average range out of a number (m) subgroups, each of size n , and $\bar{\bar{x}}$ is the *grand average*, also calculated from the same m subgroups. In Fig. 4, we give an example of applying an $\bar{X}$ - R chart for the control of the uniformity and the thickness of photoresist on silicon wafers.

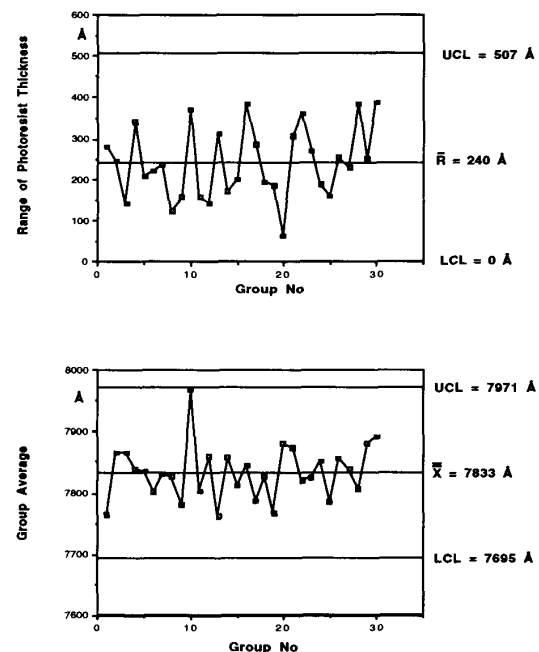


Fig. 4. $\bar{X}$ - R chart for the deposition of photoresist.

The R chart control limits for a given type I error are a function of the subgroup sample size, and can be derived from statistical tables. From these statistics we can also extract the operating characteristic function of the R chart, shown in Fig. 5, which is also adapted from [9]. Clearly, the R chart can detect with certainty only relatively large changes in the process spread.

2) *The $\bar{X}$ -S Charts:* When the subgroup size is small (less than about 5), the range of a group of measurements is a very efficient indicator of the process spread. When the subgroup size grows to more than 10, however, then the

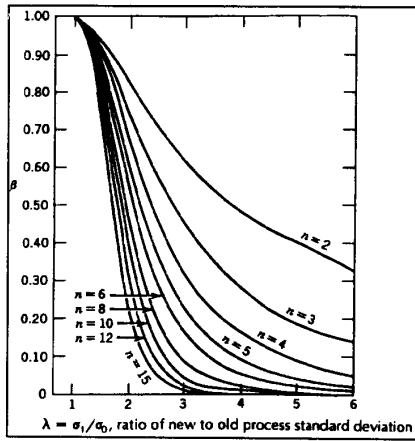


Fig. 5. Operating characteristic curves of the R chart.

familiar sum of squares is a better indicator of the process spread:

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}. \quad (10)$$

This value, along with constants which depend on the group size, is used to derive the limits of the two companion charts. The derivation is similar to that of the $\bar{X}$ - R charts.

$$CL_s = \bar{s} \pm 3 \frac{\bar{s}}{c_4} \sqrt{1 - c_4^2} \quad (11)$$

$$CL_{\bar{x}} = \bar{x} \pm \frac{3s}{c_4 \sqrt{n}}. \quad (12)$$

Here, c_4 is a proportionality factor which is needed to make σ an unbiased estimator of σ , according to the formula $E(s) = c_4 \sigma$. This factor depends on the group size n , and its value can be found in standard statistical tables, such as the ones in [9].

3) *Rational Subgrouping*: One note of caution must be inserted at this point: unless the variation that is affecting the group average (i.e., the group-to-group variation) is the same as the variation that determines the group spread (i.e., the within-the-group variation), the $\bar{X}$ limits should not be calculated from R . This means that (9) should not be used. This is a common error in semiconductor manufacturing applications, where the subgroup that looks most "natural" to the process engineer is the wafer. The causes and magnitude of the variation of a parameter *within* the wafer, however, are often very different than those of the variation of the same parameter *between* wafers. This is especially true in the case of single wafer processing, in which all points on a wafer experience a more or less uniform processing environment, although the run by run inconsistencies of the equipment often induce significant variation between wafers. Another problem might arise from deterministic radial patterns across a wafer. In this

case, (6) cannot be used to estimate the limits of the R chart. Pioneering work in processing these nonrandom spatial effects is discussed in [10].

4) *The Moving Range Chart*: Some parameters cannot be easily grouped, either because their readings are expensive, or because they are monitored continuously over a period of time. Temperature readings, for example, or readings resulting from expensive tests such as SEM measurements, cannot be easily grouped. Such parameters can be very effectively controlled by the moving range chart, a simple derivative of the $\bar{X}$ - R chart, where now the "group" is assumed to consist of two consecutive readings. In this way, group #1 consists of readings 1 and 2, group #2 includes readings 2 and 3, group #3 readings 3 and 4, etc.

The moving range chart is a powerful tool because of its simplicity. This chart permits the intuitive estimation of the in-control variance of the monitored parameter. As such, it is very useful for the control of continuously varying parameters, such as periodic temperature or pressure readings. One note of caution is necessary at this point: frequently sampled, continuously varying parameters are often *auto-correlated*, i.e., consecutive readings tend to depend on each other. Such parameters violate the IIND assumption (that readings should be *identically, independently, and normally* distributed). Even if the original parameter is IIND however, the consecutive differences will not be. The treatment of such parameters belongs to the rather advanced SPC chapter of time series analysis. The violation of the IIND assumption often results in false alarms through rules 6 and 7 of the Western Electric set as described in Section II-C.

As an example, Fig. 6 presents the application of the moving range chart for the control of real-time temperature readings from a polysilicon deposition furnace. In this example we are monitoring the temperature differential (i.e., the temperature reading at the center, minus the reading at the inlet of the reactor) via a moving range chart (Fig. 6(a)) and a chart of individual readings (Fig. 6(b)).

C. Runs Rules and the Western Electric Set

The control schemes discussed thus far deal with a very specific departure from the state of control—that of a clear shift in the value of the mean or the variance of the underlying distribution. In general, however, we are interested in seeing any sign of nonrandomness in our data. Indeed, nonrandomness may appear in many different ways, some of which do not involve a transgression of the three-sigma control limits. A popular set of runs rules is the *Western Electric* set, as summarized in Fig. 7.

Referring to part 6 of Fig. 7, consider the situation when we are recording the thickness of the photoresist layer. Although no three-sigma alarms are present, a large number of consecutive points appear to regularly alternate near the center line of the chart. This is a clear indication of nonrandomness that might result from some periodic maintenance pattern performed on the machine in question. Other such situations might lead to the stratification of

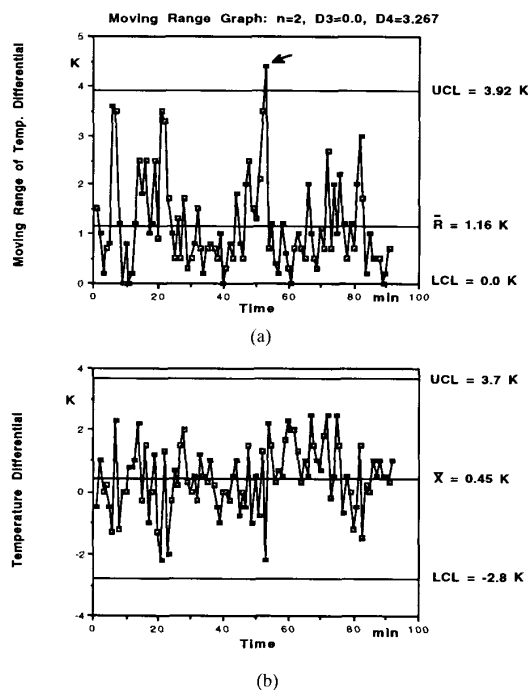


Fig. 6. A moving range chart for temperature control in a polysilicon deposition reactor. (a) The temperature differential via a moving range chart and (b) a chart of individual readings.

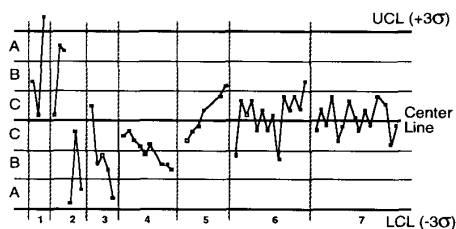


Fig. 7. Summary of the Western Electric rules.

- 1) Any point beyond three σ UCL or LCL.
- 2) 2/3 consecutive points on the same side, in zone A or beyond.
- 3) 4/5 consecutive points on the same side, in zone B or beyond.
- 4) 9/9 consecutive points on the same side of the centerline.
- 5) 6/6 consecutive points increasing or decreasing.
- 6) 14/14 consecutive points alternating up and down.
- 7) 15/15 consecutive points on either side in zone C.

points, where many consecutive points might appear in the same narrow region on the control chart.

So, in addition to the three-sigma limits, a number of "runs rules" have been introduced to identify such nonrandom situations. In general, the application of a runs rule involves the separation of the control chart into a number of zones. As an example, one such rule is used to issue an alarm whenever 4 out of 5 consecutive points fall between zero and three sigma on either the positive or the negative side of the center line. The violation of this rule is depicted in part 2 of Fig. 7.

In general, the application of multiple runs rules complicates the evaluation of the types I and II risks. A number

of complex simulators have been written to analyze the risks and the ARL of charts employing general sets of runs rules, and some are described in [12]. Runs rules have also been designed so that they optimize the cost effectiveness of a chart, taking into account the cost of each types I and II occurrence [9], [13]. Most of these schemes offer little intuition to the non-statistician, and they have not been used to a significant extent by the semiconductor manufacturing industry.

D. Controlling Defect Counts—An Example Using the Poisson Model

Many high complexity VLSI and ULSI products are vulnerable to defects that land on the wafer during processing. Consequently, ever since the inception of Integrated Circuits, a number of so-called *yield models* have appeared in the literature and have been used extensively by process engineers [14], [15]. The objective of these models is to predict the yield of a new IC design, given the defect density, the design rules, the die size, etc. Most of these models assume that the defect density is either constant, or that it obeys a known, stationary statistical distribution.

Once a new IC reaches production, however, it is usually accompanied by some modifications in the technology. This means that a new IC product usually starts at a low yield and it follows a "yield transient" while the process engineers learn the new process. Once an acceptable yield has been established, it must be monitored in order to ensure that the defect generation mechanism of the process remains under control. Another reason for statistical monitoring of the yield is to identify and quantify any yield changes that follow process modifications.

A distinct family of control charts, known as *attribute charts*, may be used in the control of process attributes such as the fraction of nonconforming die and the respective defect counts. These charts are based on statistical models that describe the particle generating mechanisms during processing. Although these types of charts are very simple, they directly monitor the fabrication line yield, which is, after all, one of the most important characteristics of a high volume production line.

The most direct method for monitoring yield is the direct application of the *fraction nonconforming* chart, also known as the *P chart*. In order to create such a chart we need to derive a relevant statistical model of the production process. This model is based on the assumptions that: a) the process is operating without any assignable causes, and b) each die has a constant probability p of being defective.

Under these assumptions, if we sample n die at a time, the probability that we will find x defectives ($P\{D = x\}$) is given by the binomial distribution:

$$P\{D = x\} = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 1, 2, \dots, n. \quad (13)$$

If we measure the proportion of defective die from multiple groups (lots) and use the average as the monitored statistic, the mean is equal to the probability of failure p and the variance is also known. More specifically, if we count the

defective die (D) out of a group of n die, and if we use m of these groups to establish the control chart, then the centerline of the chart is given by

$$\bar{p} = \frac{\sum_{i=1}^m \hat{p}_i}{m} \quad (14)$$

where the estimated fraction nonconforming for each group is given by

$$\hat{p}_j = \frac{D_j}{n}, \quad j = 1, 2, \dots, n. \quad (15)$$

Finally, the three-sigma control limits are given by (16) below, assuming that the sample size n is large enough so that the binomial distribution is almost symmetrical about its mean. This implies that it can be approximated by a normal distribution. In this case the control limits are given by

$$CL = \bar{p} \pm 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}. \quad (16)$$

There are several rules which deal with the assumption of symmetry and with the design of the P chart in general. These rules are described in detail in [9].

Similar charts can be used to monitor the number of defects when assuming a known, constant defect density c and a mechanism that generates defects according to a Poisson distribution. This control chart is known as the C chart, and its three-sigma control limits are given around the known defect density c . This defect density represents the average number of defects on each inspection unit, which might be a die, a wafer, or a batch of wafers.

$$CL = c \pm 3\sqrt{c}. \quad (17)$$

Another useful attribute chart is the U chart, which deals with the average defect count over a group of n entities such as die, wafers, or wafer batches. The control limits of the U chart are based on averaging the Poisson-distributed defect counts. Thanks to the Central Limit Theorem, this average will tend to be distributed according to a Gaussian distribution. Therefore, the three-sigma limits of the U chart are given by

$$CL = \bar{u} \pm 3\sqrt{\frac{\bar{u}}{n}} \quad (18)$$

where $\bar{u}$ is the observed average defect density over n inspection units.

The P , C , and U charts have traditionally been based on a model that describes the random generation of defects according to a Poisson distribution. The Poisson-based model, however, is unable to describe the clustering effect which causes defects to appear in groups on some of the larger IC products. Recently, more sophisticated models have been proposed to extend the application of formal control schemes to cases that cannot be modeled with a Poisson distribution [16].

The P chart example that follows in Fig. 8(a) uses data from [15], and monitors the number of defective die on

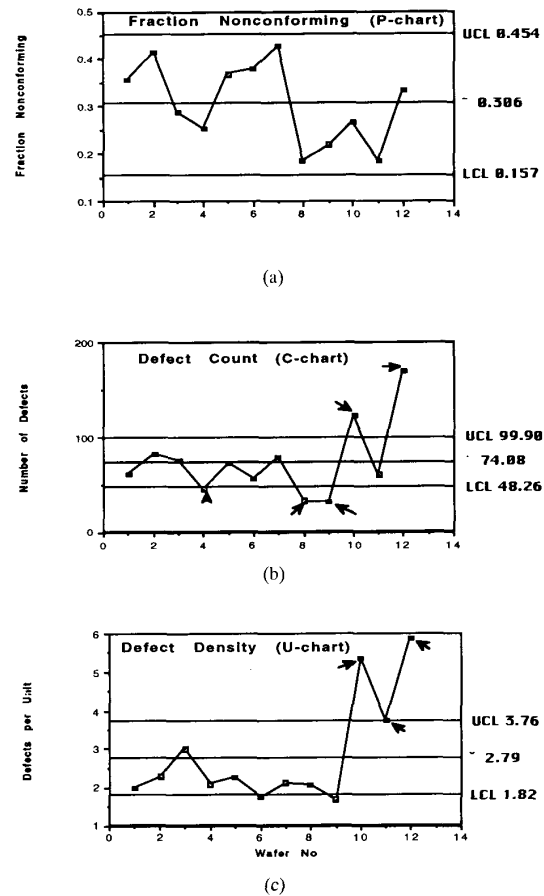


Fig. 8. (a) The P chart, built to monitor the portion of bad die per wafer. (b) The C chart, used to monitor defects/wafer. Limits are based on the Poisson Model. (c) The U chart, used to monitor the average number of defects per die on each wafer.

each wafer, one wafer at a time. Two additional charts can be applied here to help focus not so much on the IC product (wafer) but on the *causes* behind the yield fluctuation. The C chart in Fig. 8(b) shows the number of defects on each wafer, assuming that the defects are generated according to an unclustered Poisson distribution. The U chart in Fig. 8(c) monitors the average number of defects/defective die for each wafer. It is also based on unclustered Poisson statistics. Although these tools cannot account for clustering, they are powerful, straightforward tools that can successfully detect yield fluctuations.

E. Maximum Likelihood Estimation Control—The CUSUM Chart

A newer class of control charts, introduced in the late 1950's, is based on the concept of *maximum likelihood*. These charts use the cumulative sum (CUSUM) of process deviations in order to generate an alarm [17]. The approach is quite sensitive to small, persistent deviations of a process, such as those due to subtle miscalibrations or small changes in the quality of incoming material. Since most semiconduc-

tor processes are well instrumented against large deviations, CUSUM schemes can effectively capture the remaining small deviations and are very suitable to semiconductor process control. Here, the monitored statistic is equivalent to the accumulated deviation of the recorded parameter from its target:

$$\sum_{i=1}^n (\bar{u} - \mu_0). \quad (19)$$

The formulas necessary to produce a chart based on this statistic are given below:

$$d = \left(\frac{2}{\delta^2} \right) \ln \left(\frac{1 - \beta}{\alpha} \right) \quad (20)$$

$$\theta = \arctan \left(\frac{\Delta}{2A} \right) \quad (21)$$

$$\delta = \frac{\Delta}{\sigma_{\bar{x}}} \quad (22)$$

$$d = -\frac{2 \ln \alpha}{\delta^2}. \quad (23)$$

Here, d is the lead distance (in number of samples) and θ is the angle of the V-shaped limits. The type I error of this chart is α . δ is defined as the deviation to be detected with a type II error β . The same deviation, expressed in number of sigmas of the sampling average, is δ . Finally, a scaling factor is needed to relate the vertical to the horizontal scales in the graph, so that the angle of the V-shaped limits is correctly drawn. This scaling factor is A and it is usually given values between 1 and 2 s , where s is the estimated standard deviation of x . An example of the application of the CUSUM chart is shown in Fig. 9.

Due to the inherent smoothing of the CUSUM chart, (the integration acts as a low-pass filter that effectively eliminates any spikes), this scheme is ideal for automatic feedback control applications.³ This way meaningful long term changes can be observed and compensated separately from unique disturbances.

Another example of the application of the CUSUM chart in semiconductor manufacturing is drawn from the run by run control of a photolithographic workcell. In this application we are interested in maintaining consistent levels of photoactive compound concentration within the photoresist layer. The level of concentration can be inferred by means of a specialized reflectance measurement [19]. Any change in the measured reflectance can lead to assignable causes of variation in the consistency of our photoresist supply or in the consistency of operation of our spin/coat and bake equipment. As can be seen in the accompanying figures, the CUSUM chart in Fig. 10 responds faster and gives a

³Another chart with inherent smoothing capabilities is the exponentially weighted moving average control scheme (EWMA) [18]. Although not yet popular in semiconductor manufacturing, this scheme offers a reasonable compromise between the large shift responsiveness of the Shewhart chart and the small shift sensitivity of the CUSUM chart.

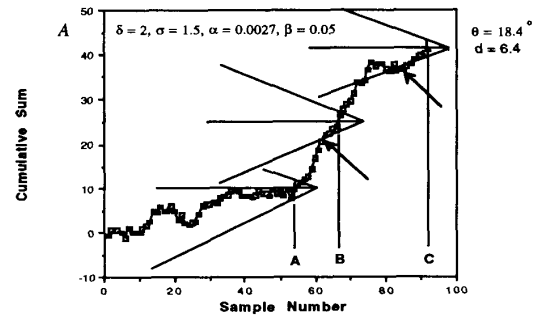


Fig. 9. CUSUM chart for temperature control during poly deposition.

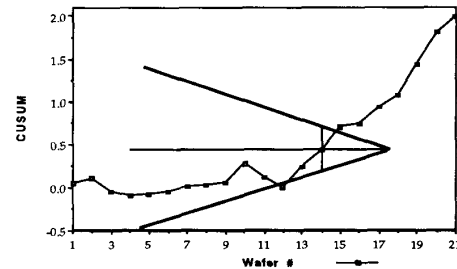


Fig. 10. CUSUM chart of measured photoresist reflectance.

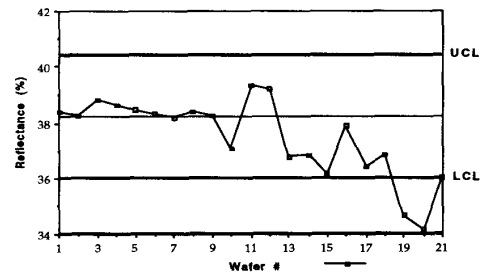


Fig. 11. Shewhart chart of the photoresist reflectance pictured in Fig. 10.

more unambiguous picture of the Shewhart chart in Fig. 11. In this case, the assignable cause has been found to be a miscalibration of the prebake temperature.

III. NOVEL SPC METHODS IN SEMICONDUCTOR MANUFACTURING

Although SPC has been applied on high volume production since the early 1930's, the original techniques have evolved significantly over the years in order to accommodate the needs of changing manufacturing technology. A major force behind the evolution of statistical process control is the recent availability of automated *in situ* data collection and real-time data processing capabilities. As a result, comprehensive control schemes which would have been impractical two decades ago are now finding their way

onto the factory floor. This revolution is likely to have a major impact on semiconductor manufacturing.

The special control requirements of semiconductor manufacturing stem from the poor repeatability of several critical VLSI manufacturing steps and also from the need to achieve high *process capability*⁴ with technologies that have little time to mature before they are applied in production. This situation becomes even more complicated when typical production runs between recipe changes are short. In addition, in order to achieve high reliability of production, several critical processes should be monitored by means of multiple real-time parameters. Unfortunately, these parameters are typically cross-correlated and non-IIND.⁵

These circumstances create special requirements and opportunities for the application of SPC in semiconductor manufacturing. In the rest of this paper we will describe some of these special techniques, including multivariate, model-based and real-time applications of statistical process control in semiconductor manufacturing.

A. Multivariate Control—Hotellings T^2 Chart

Often, a critical processing step might be monitored by means of recording several parameters. One example of this is the monitoring of dry polysilicon etching through the etch rate, etch uniformity, selectivity to photoresist and selectivity to oxide. Although these measurements carry important information about the process, they also need special SPC schemes for their analysis. More specifically, an important consideration is the fact that such parameters are very likely to be statistically correlated with each other. This means that if we use a number of independent control charts, the overall manufacturer's (α) and consumer's (β) risks cannot be evaluated correctly.

In response to this problem, several multivariate control techniques have emerged and are in use today. These schemes alert the operator to changes in the mean vector or the covariance matrix of a group of controlled parameters. One of the most popular multivariate control schemes is based on Hotelling's T^2 statistic. This statistic, defined below, is sensitive to the collective deviations of a number of cross-correlated IIND parameters from their respective targets. Assuming that we have p such parameters whose variance-covariance matrix is known and does not change (even if the process goes out of control), the T^2 statistic is given by the formula:

$$T^2 = n(\bar{\mathbf{x}} - \boldsymbol{\mu})^T \mathbf{S}^{-1} (\bar{\mathbf{x}} - \boldsymbol{\mu}) \quad (24)$$

where n is the size of each measurement subgroup, $\bar{\mathbf{x}}$ is the vector of the group averages as measured, $\boldsymbol{\mu}$ is the vector

⁴The *process capability* measures of C_p (used when a process is centered around its specifications) and C_{pk} (for skewed processes) are related to how suitable is a process for the application at hand. C_p is defined as the ratio of the specification window over the six-sigma spread of the process, while C_{pk} is similarly defined for potentially skewed processes.

⁵Identically, independently, and normally distributed. See Section II-A.

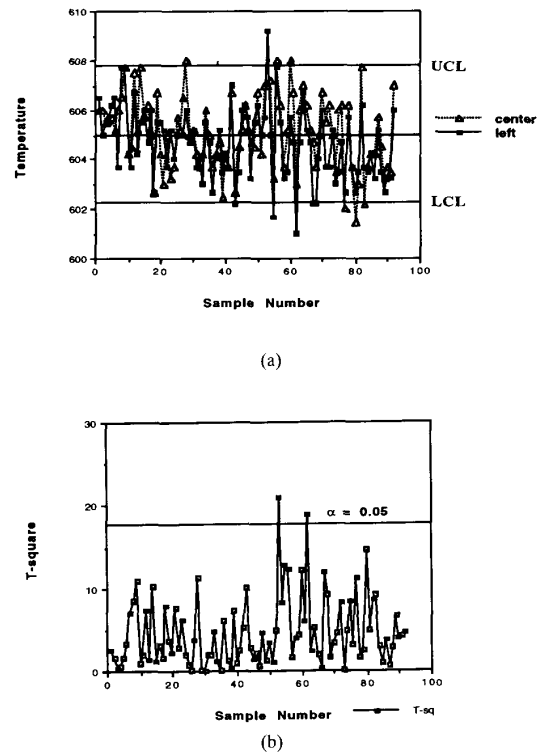


Fig. 12. (a) Center and left temperature averages (4 readings per group) in LPCVD furnace. (b) T^2 plot for center and left temperature average.

of the group means (target values), and $\mathbf{S}$ is the $p \times p$ covariance matrix. The superscript (T) is used to indicate the transpose operation. All the vectors are originally defined as $1 \times p$ arrays (i.e., columns). Under the assumption that, when under control, all the random variables are identically and independently, normally distributed (IIND) around their respective mean μ_i , then the α -level upper control limit of this one-sided chart is given with the help of the chi square distribution:

$$UCL = \chi^2_{\alpha, p}. \quad (25)$$

If the parameter mean and the covariance matrix have been estimated from a small number of samples, then the upper control limit is more correctly defined with the help of the F distribution [26].

An example of the T^2 statistic is shown in Fig. 12(a) and (b). Here we use two temperature readings at either end of an LPCVD deposition tube in order to monitor the temperature during the deposition of critical polysilicon films. In Fig. 12(a) we present two control charts with limits set for $\alpha = 0.05$. Since the process was under control, we would only expect to see about five false alarms, yet we see many more. In Fig. 12(b) the one-sided control limit has also been set for $\alpha = 0.05$, but now we only receive two false alarms. Obviously, the T^2 statistic presents a far clearer picture of the process status and is much less likely to introduce false alarms.

B. Model-Based Statistical Process Control—The Regression Chart

Semiconductor manufacturing equipment is often used to support an array of products, each requiring a different processing recipe. According to traditional SPC practice however, multiple recipes (or intentional changes of any kind) cannot be present if a control chart is to be used. This requirement is very restrictive in semiconductor manufacturing, where change is the rule and long runs are the exception. A variant of SPC, known as *model-based SPC* can be used to solve this problem.

The foundation of model-based SPC is the Regression Chart [21], introduced by Mandel in 1969.⁶ Instrumental in the regression chart is the use of a regression model that predicts the nominal response of the various equipment as a function of their settings. The *residual* of this response is obtained as the difference between the predicted and the observed equipment response. Since the statistics of the residual are well known, a Shewhart control chart can be used to control it. Out of control points are then treated as indications of assignable causes.

An important special application of the regression chart concerns points that appear to be *systematically* out of control. This usually means that the equipment has drifted and that the models must be reevaluated. This situation can be detected with the help of the cumulative student-*t* statistic. In this way, an *adaptive* statistical process control scheme can be implemented so that abrupt changes can be detected, even as the equipment model is being continuously adapted. This model can then be used for further process optimization.

An example of such an application appears in Fig. 13(a), where the low-pressure chemical vapor deposition tube is being controlled by a regression chart. The regression equation has been built to predict the deposition rate as a function of temperature, SiH_4 flow and pressure [12], [24]. The control limits reflect both the experimental error of the tube as well as the prediction error of the regression model. The situation depicted in Fig. 13 shows that the tube is out of control, since several of the plotted points violate the control limits. In the first graph in Fig. 13 the regression model is also significantly different than the equipment response. In the second graph the model has been *adapted* (recentered), yet this adaptation did not interfere with the primary detection of one of a kind assignable causes. Similar schemes are now in use for the control of experimental photolithographic operations in the Berkeley Microfabrication Laboratory. The detailed description of the model-based SPC scheme will be the subject of a future publication.

C. Time Series Analysis

Modern semiconductor manufacturing equipment are outfitted with sensors capable of monitoring a number of

⁶The term *model-based SPC* has recently been reintroduced in [25] to describe a similar concept where the model residuals are applied on a Shewhart chart. This simplified scheme does not take into account the prediction error of the regression equation.

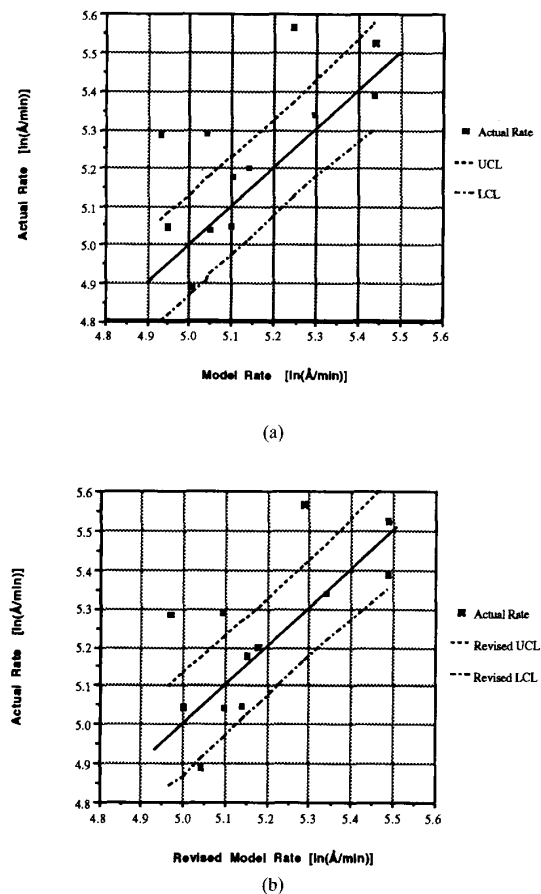


Fig. 13. (a) Model-based SPC on an LPCVD tube. Systematic error indicates chronic drift. (b) Model-based SPC on LPCVD tube. Model has been adapted to account for chronic drift.

critical process parameters such as temperature, pressure, gas flows etc. In addition, most new equipment can automatically upload these readings to a host computer system with the help of the SECSII [20], the standard interequipment communication protocol instituted by SEMI.

A problem that often arises in conjunction with such rapid, continuous parameter readings is that each new value tends to be *statistically related* to previously measured values. The existence of autocorrelation in the controlled parameters violates one of the most basic assumptions behind the design of standard SPC schemes, namely that all samples are *IIND* random variables. In order to cope with this problem, the monitored parameter might be modeled by means of an appropriate *time series* model. Time series models, such as the well known autoregressive integrated moving average (ARIMA), can be used to forecast each measurement and deduce the *forecasting error* [22]. This error can then be assumed to be an independently distributed random variable, and it can be used with traditional SPC schemes. An example of this application appears in Fig. 14. In this example, we examine the real-time readings

of temperature collected from a low pressure, chemical vapor deposition tube. In Fig. 14(b) we see significant autocorrelation between temperature readings separated by one time period. Based on this autocorrelation, a model is generated to *forecast* each new reading. Finally, in Fig. 14(c) the forecasting error is treated as an IIND parameter. Compare Fig. 14(a) to Fig. 14(c) and notice the appreciable reduction of alarms. The limits are set for $\alpha = 0.05$. The model used in this simple example is the ARIMA (1,0,0) model since it involves one autoregressive term, no differentiation and no moving average terms.

The power of time series models becomes apparent when one considers the recent popularity of multichamber equipment [27]. These so-called "cluster" tools promise much improved process quality by automatically sequencing a wafer through several processing steps without exposing it to the cleanroom atmosphere during transport. This, however, makes it impossible for the operator to inspect the wafers between steps in order to make sure that individual operations remain under control. The only way to achieve this is through the statistical monitoring of real time sensor readings, and this is only possible with the help of the appropriate time series filtering [28].

IV. COMPUTER-INTEGRATED MANUFACTURING AND SPC

Easy access to information, both for generating an alarm as well as for discovering its assignable cause, is instrumental in SPC operations. With current computer technologies, it is possible to construct a physically distributed but logically integrated database. This will greatly facilitate data manipulation across the manufacturing floor and will lead to high productivity.

Four recent advances contribute to this realization. The most important advance is the development of relational database systems which reduce the effort required for both the initial development and the subsequent maintenance and modification of a system. This is because relational databases support interfaces which allow end-users (in this case process, maintenance, and yield engineers) to easily manipulate the information stored in the database.

The second major development has been the industry-wide acceptance of high bandwidth communication standards (local area networks, or LAN's) for linking systems from different vendors. LAN's make it possible to connect process control applications directly to the fabrication equipment. Consequently, in-line and in-process measurements can be automatically collected and analyzed.

A third important development is the emergence of distributed database management systems. Due to distributed database systems, information is physically stored at many nodes yet appears to the user as a coherent entity. The distributed database system determines where the data is located, generates an efficient plan to retrieve or update it, and ensures its consistency and integrity.

The fourth important element is the spreading use of artificial intelligence technologies. Many knowledge intensive, error-prone activities in semiconductor manufacturing

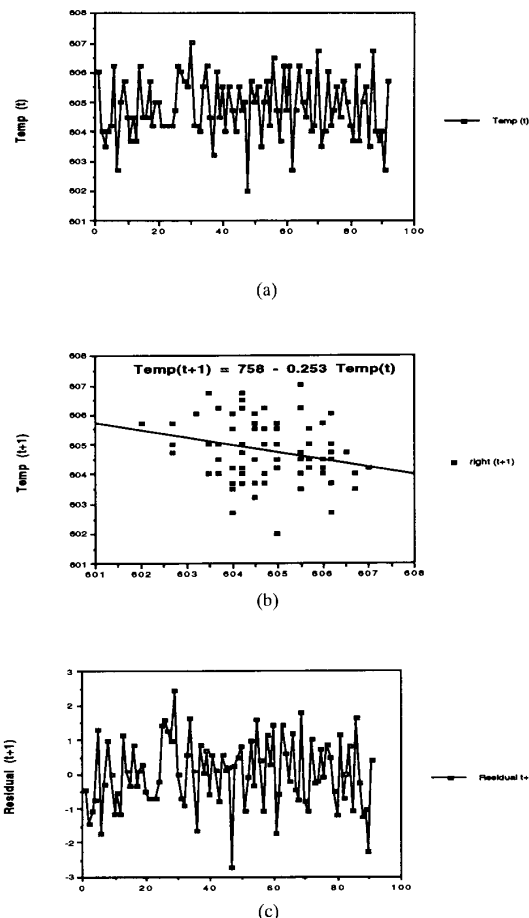


Fig. 14. (a) Filtering real-time non-IIND data for SPC. "Raw" temperature readings over time. $\alpha = 0.0027$. (b) Filtering real-time non-IIND data for SPC. Autocorrelation in temperature readings. (c) Filtering real-time non-IIND data for SPC. Chart of IIND residuals from ARIMA (1, 0, 0).

can be automated by the use of AI techniques. The need for automated decisions in planning, scheduling, diagnosis, maintenance, etc. becomes even more pressing in view of the complexities of the new submicron ULSI processes. The objective of the Berkeley CIM architecture is to develop software modules for controlling VLSI processing steps, and to demonstrate a flexible framework for combining these modules into an integrated CIM system.

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- [1] D. A. Hounshell, *From the American System to Mass Production, 1800-1932: The Development of Manufacturing Technology in the United States*. Baltimore, MD: The Johns Hopkins Univ. Press, 1984.
- [2] R. Jaikumar, "From filing and fitting to flexible manufacturing: A study in the evolution of process control," *Working Paper*, Feb. 1988.
- [3] J. Gies, "Automating the worker," *The American Heritage of Invention and Technology*, vol. 6, no. 3, pp. 56-64, Winter 1991.
- [4] P. Drucker, "The emerging theory of manufacturing," *Harvard Business Rev.*, May-June 1990.
- [5] G. P. Box, W. G. Hunter, and J. S. Hunter, *Statistics for Experimenters*. New York: Wiley-Interscience, 1978.
- [6] H. Kume, "Statistical methods for quality improvement," *Ass. Overseas Technical Scholarship*, 1985.
- [7] K. Ishikawa, "Guide to quality control," *Asian Productivity Organization—Quality Resources*, 1982.
- [8] G. Taguchi, E. Elsayed, and T. Hsiang, *Quality Engineering in Production Systems*. New York: McGraw-Hill, 1989.
- [9] D. C. Montgomery, *Introduction to Statistical Quality Control*. New York: Wiley, 1991, 2nd ed.
- [10] J. K. Kibarian, "Statistical Diagnosis of IC Process Faults," Ph.D. dissertation, Research Rep. CMUCAD-90-52, Carnegie Mellon University, Pittsburgh, PA, Dec. 1990.
- [11] J. B. Keats and N. F. Hubele, Eds., *Statistical Process Control in Automated Manufacturing*. New York: Marcel Dekker, 1989.
- [12] C. J. Spanos, "Special Issues in Semiconductor Manufacturing—I," Electronics Research Lab. M90/8 EECS, Univ. California, Berkeley, CA, Jan. 1990.
- [13] —, "Special Issues in Semiconductor Manufacturing—II," Electronics Research Laboratory M91/8 EECS, Univ. California, Berkeley, CA, Jan. 1991.
- [14] C. H. Stapper, "Fact and fiction in yield modeling," *Microelectronics J.*, vol. 20, no. 1-2, pp. 129-151, Jan. 1989.
- [15] J. A. Cunningham, "The use and evaluation of yield models in integrated circuit manufacturing," *IEEE Trans. Semiconductor Manufac.*, vol. 3, pp. 60-71, May 1990.
- [16] D. Friedman and S. Albin, "Clustered defects in IC fabrication: Impact on process control charts," *IEEE Trans. Semiconductor Manufac.*, vol. 4, pp. 36-42, Feb. 1991.
- [17] E. S. Page, "Continuous inspection schemes," *Biometrika*, vol. 41, pp. 100-115, 1954.
- [18] J. M. Lucas and M. S. Saccucci, "Exponentially weighted moving average control schemes: Properties and enhancements," *Technometrics*, vol. 32, no. 1, pp. 1-12, Feb. 1990.
- [19] Z.-M. Ling, S. Leang, and C. J. Spanos, "In-line supervisory control in a photolithography workcell," presented at the SPIE Symp. on Microelectronics Processing Integration, Santa Clara, CA, Oct. 1990.
- [20] Semiconductor Equipment and Materials International Standards E84, Semiconductor International, Mountain View, CA, 1984.
- [21] B. J. Mandel, "The Regression Control Chart," *J. Quality Technol.*, vol. 1, no. 1, pp. 1-9, Jan. 1969.
- [22] G. E. P. Box and G. M. Jenkins, *Time Series Analysis: Forecasting and Control*. San Francisco, CA: Holden-Day, 1976, 2nd ed.
- [23] D. A. Hodges, L. A. Rowe, and C. J. Spanos, "Computer Integrated Manufacturing," presented at the Int. Electronics Manufacturing Technology Symp., San Francisco CA, Sept. 1989.
- [24] K.-K. Lin and C. J. Spanos, "Statistical modeling of semiconductor manufacturing equipment: An Application for LPCVD," *IEEE Trans. Semiconductor Manufac.*, vol. 3, pp. 216-229, Nov. 1990.
- [25] E. Sachs, R.-S. Guo, S. Ha, and A. Hu, "Process control system for VLSI fabrication," *IEEE Trans. Semiconductor Manufac.*, vol. 4, pp. 134-144, May 1991.
- [26] Richard J. Harris, *A Primer of Multivariate Statistics*. New York: Academic, 1975.
- [27] K. Shankar, "Cluster Tools: A \$2.2 Billion Market by 1994," *Solid State Technol.*, vol. 33, no. 10, p. 43, Oct. 1990.
- [28] C. J. Spanos, H. Guo, A. Miller, and J. Levine-Parril, "Real-time SPC using tool data," *IEEE Trans. Semiconductor Manufac.*, vol. 5, Nov. 1992.



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